C++ is standardized by the International Organization for Standardization (ISO). The latest standard version published in December 2017 as ISO/IEC 14882:2017 which is informally known as C++17. The first standardized version was published in 1998 as ISO/IEC 14882:1998 which was then enhanced by the C++03 (ISO/IEC 14882:2003), C++11 (ISO/IEC 14882:2011) and C++14 (ISO/IEC 14882:2014). The Next standard version will be C++20 in 2020.

C# (C-Sharp), D, Java and newer versions of C languages has been influenced by C++.

### Benefits of learning C++

- C++ is a highly portable language and is often the language of choice for multi-device, multi-platform app development.
- C++ is an object-oriented programming language and includes **classes**, **inheritance**, **polymorphism**, **data abstraction** and **encapsulation**.
- C++ has a rich function library.
- C++ allows exception handling, inheritance and function overloading which are not possible in C.
- C++ is a powerful, efficient and fast language. It finds a wide range of applications – from GUI applications to 3D graphics for games to real-time mathematical simulations.

## 9.2 Character set

Character set is a set of characters which are allowed to write a C++ program. A character represents any alphabet, number or any other symbol (special characters) mostly available in the keyboard. C++ accepts the following characters.

Most of the Character set, Tokens and expressions are very common to C based programming languages like C, C++, Java, PHP etc.,

| | |
|---|---|
| Alphabets | A …. Z, a ….. z |
| Numeric | 0 …. 9 |
| Special Characters | + - * / ~ ! @ # $ % ^& [ ] ( ) { } = >< _ \ \| ? . , : ‘ “ ; |
| White space | Blank space, Horizontal tab (→), Carriage return (↵), Newline, Form feed |
| Other characters | C++ can process any of the 256 ASCII characters as data. |

## 9.3 Lexical Units (Tokens):

C++ program statements are constructed by many different small elements such as commands, variables, constants and many more symbols called as operators and punctuators. These individual elements are collectively called as **Lexical units** or **Lexical elements** or **Tokens**. C++ has the following tokens:

- Keywords
- Literals
- Punctuators
- Identifiers
- Operators